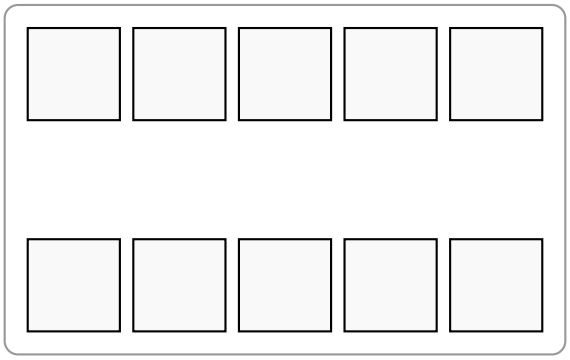
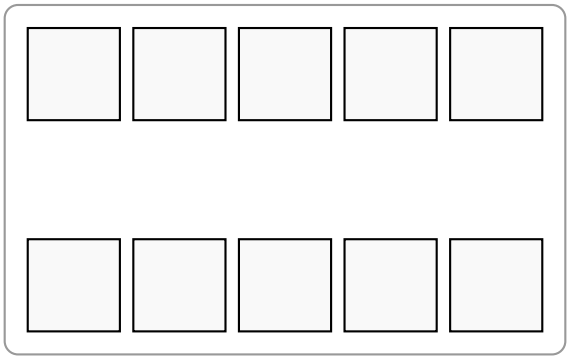
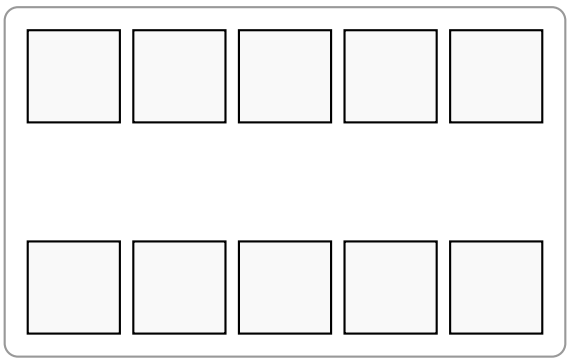
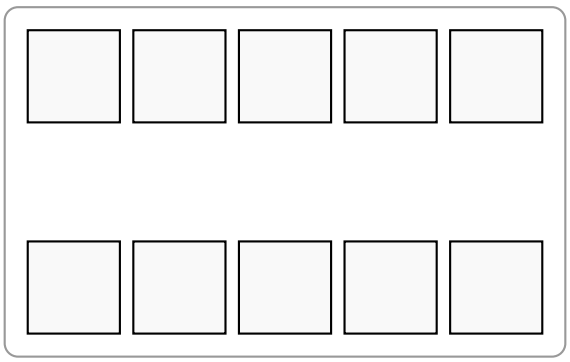
One turn: move every cube down its own arrow, then slide the whole bottom row straight up into the top row. When you draw a machine, give every top slot one arrow going out and every bottom slot one arrow coming in. A cube is home when it is back under its own picture.

Problem 5: The machines you drew in Problem 4 undo the machines above them. A machine undoes itself if two turns of it put every cube home. Draw four different machines with 5 slots that undo themselves. How can you tell from its loops whether a machine undoes itself? Explain.

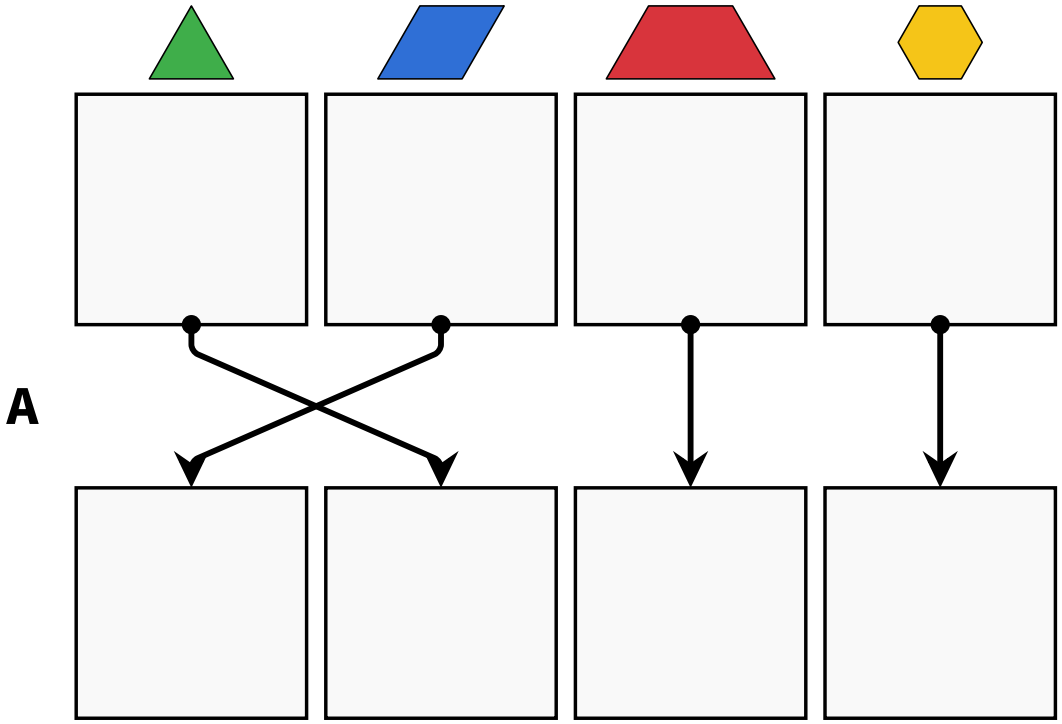








Problem 6: Put each cube under its picture on machine A and do one turn. Move the row to machine B without changing its order, and do one turn there. Draw the cubes where they end up in the row marked A then B. Then start again with one turn of B followed by one turn of A, and draw the row marked B then A. Do this for both pairs of machines. For which pair are the two rows the same? Explain why.

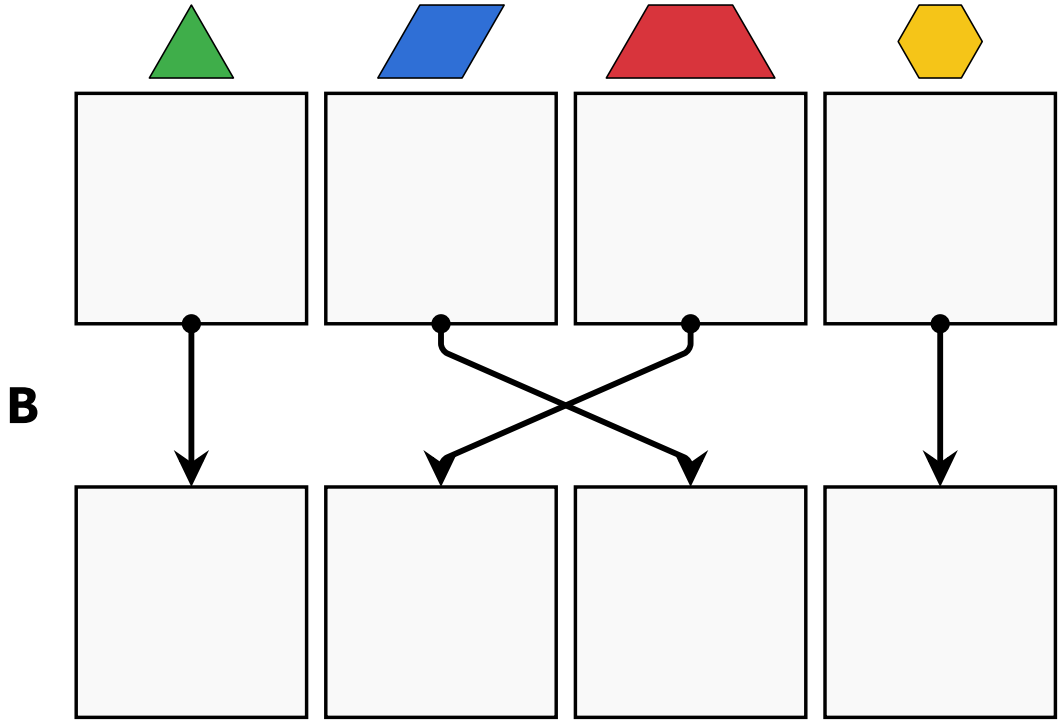


A then B

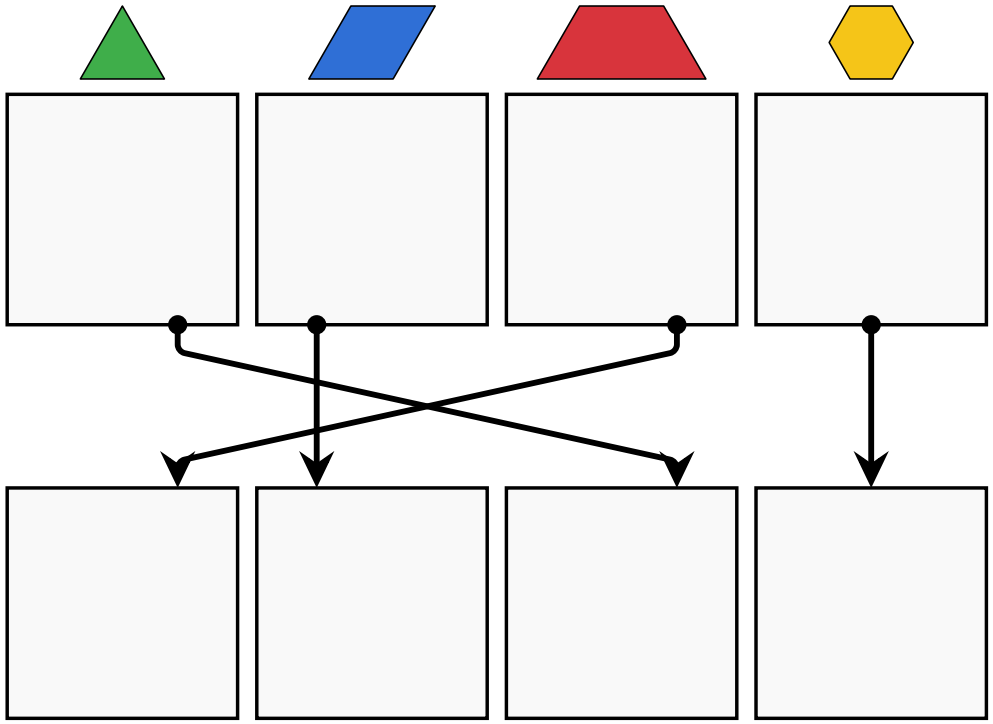


B then A





A



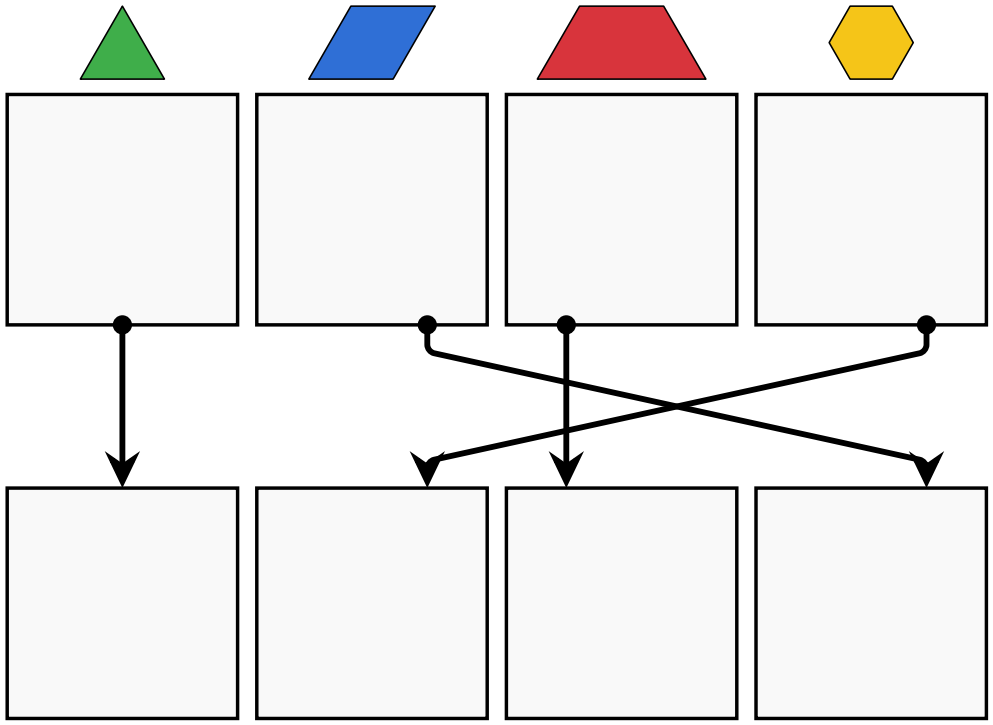
A then B



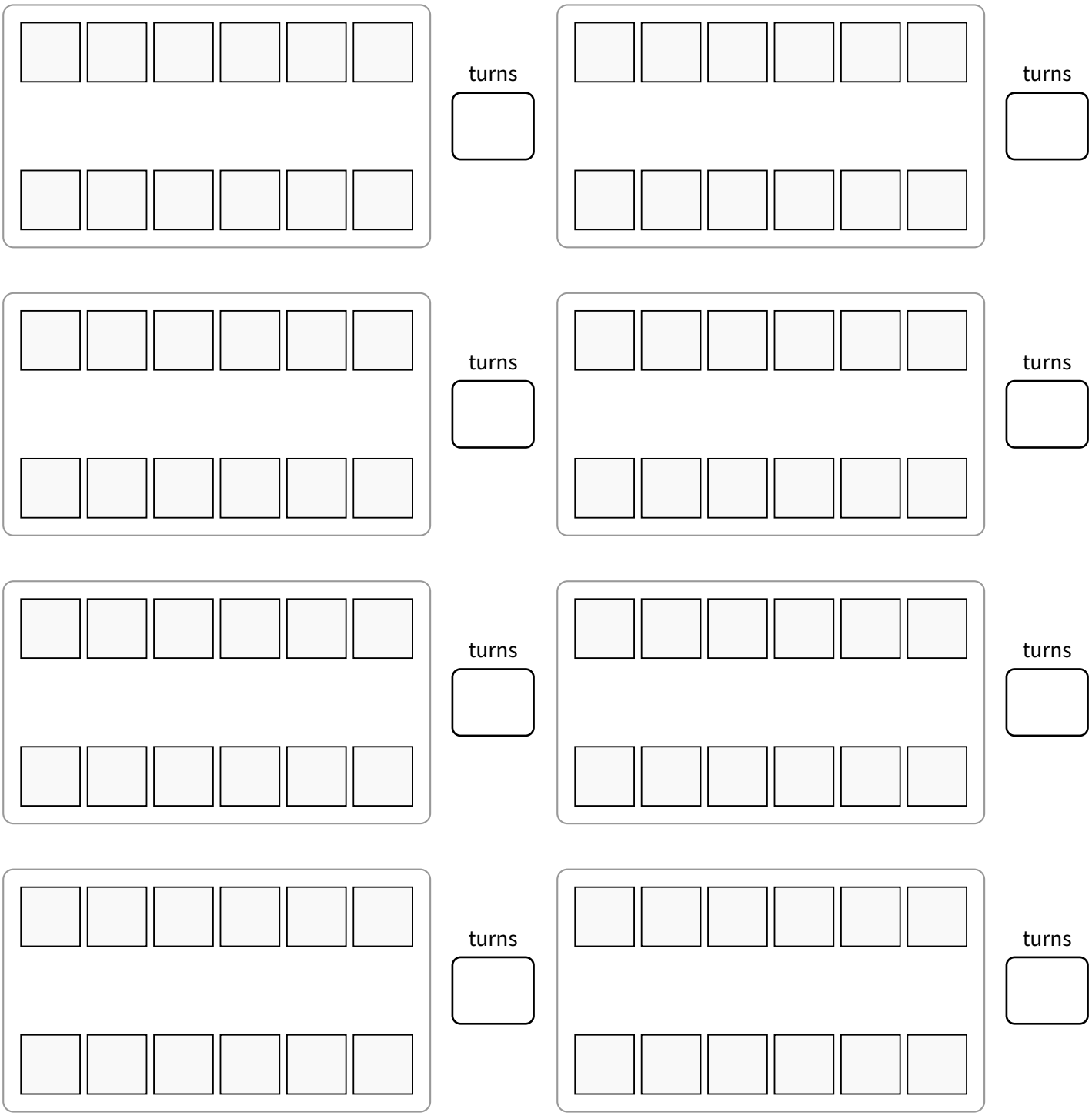
B then A



B



Problem 7: Which numbers of turns can a machine with 6 slots take to bring all the cubes home? Draw a machine for each number you find, and explain why no other number is possible.



Problem 8: Draw a machine with 7 slots that takes more than 10 turns to bring all the cubes home. What is the most turns a machine with 7 slots can take? Explain.

